Bubble sort:

```
for( i=0; i<=4-2;i++)
{
for( j=0;  j<=4-i-2;j++)
{   L      R
if(a[j]>a[j+1])
{
t=a[j];a[j]=a[j+1];a[j+1]=t;
}
}
}
```

9 3 7 -4

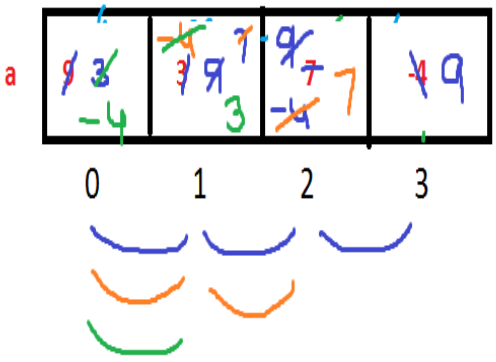
3 9 7 -4

3 7 9 -4

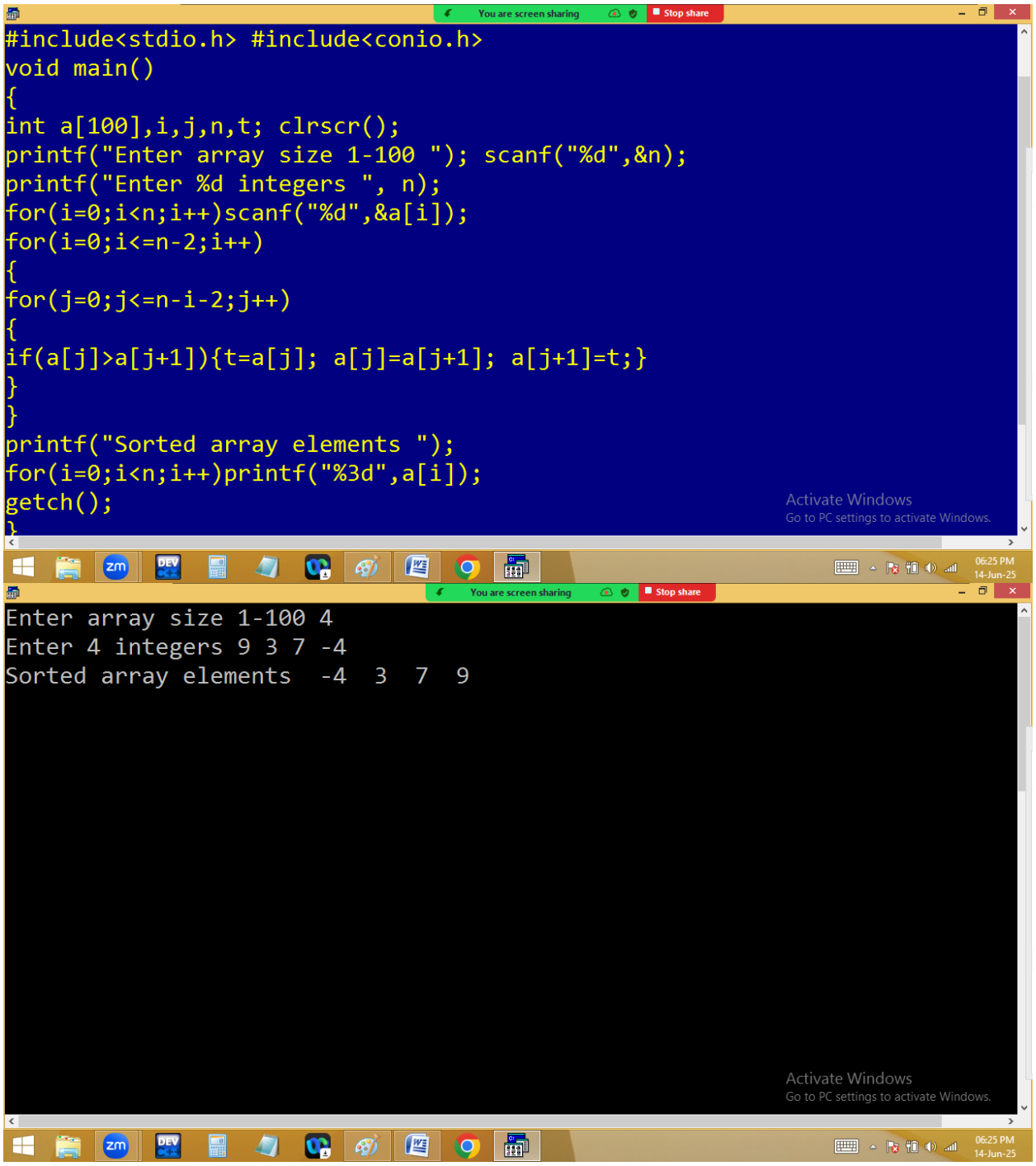
3 7 -4 9

3 -4 7 9

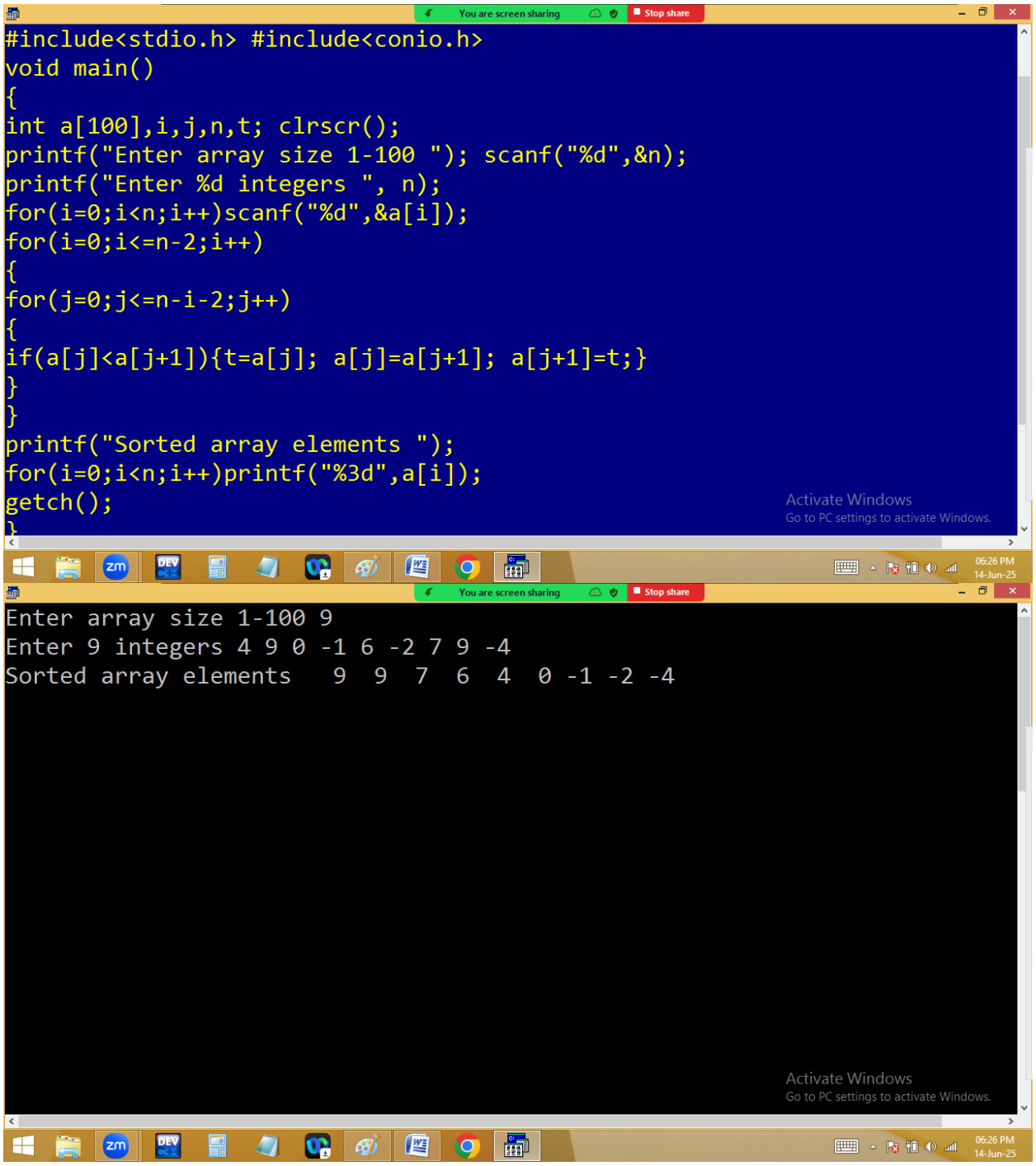
-4 3 7 9



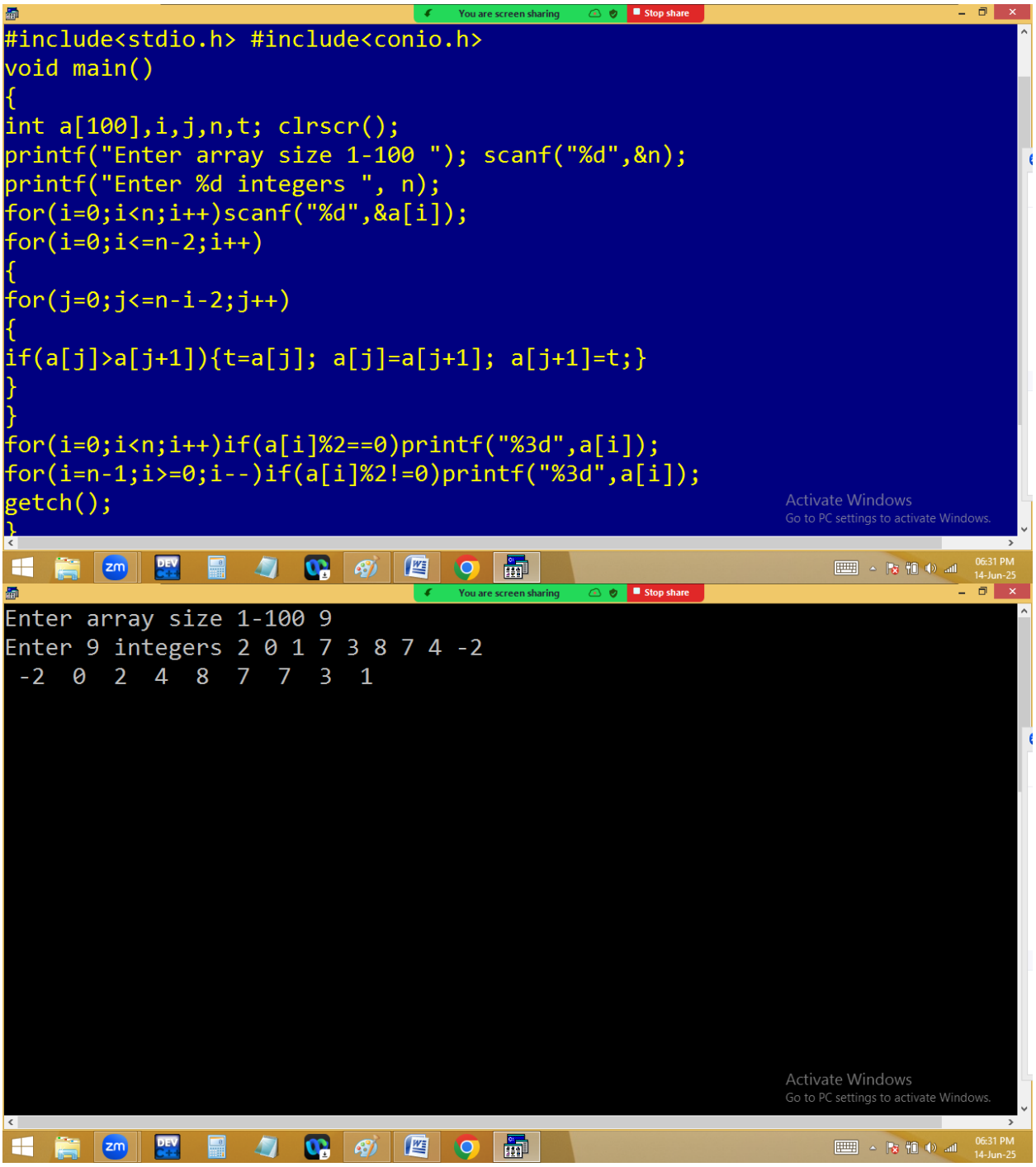
i	j	j+1
0	0-1, 1-2, 2-3	
1	0-1, 1-2	
2	0-1	



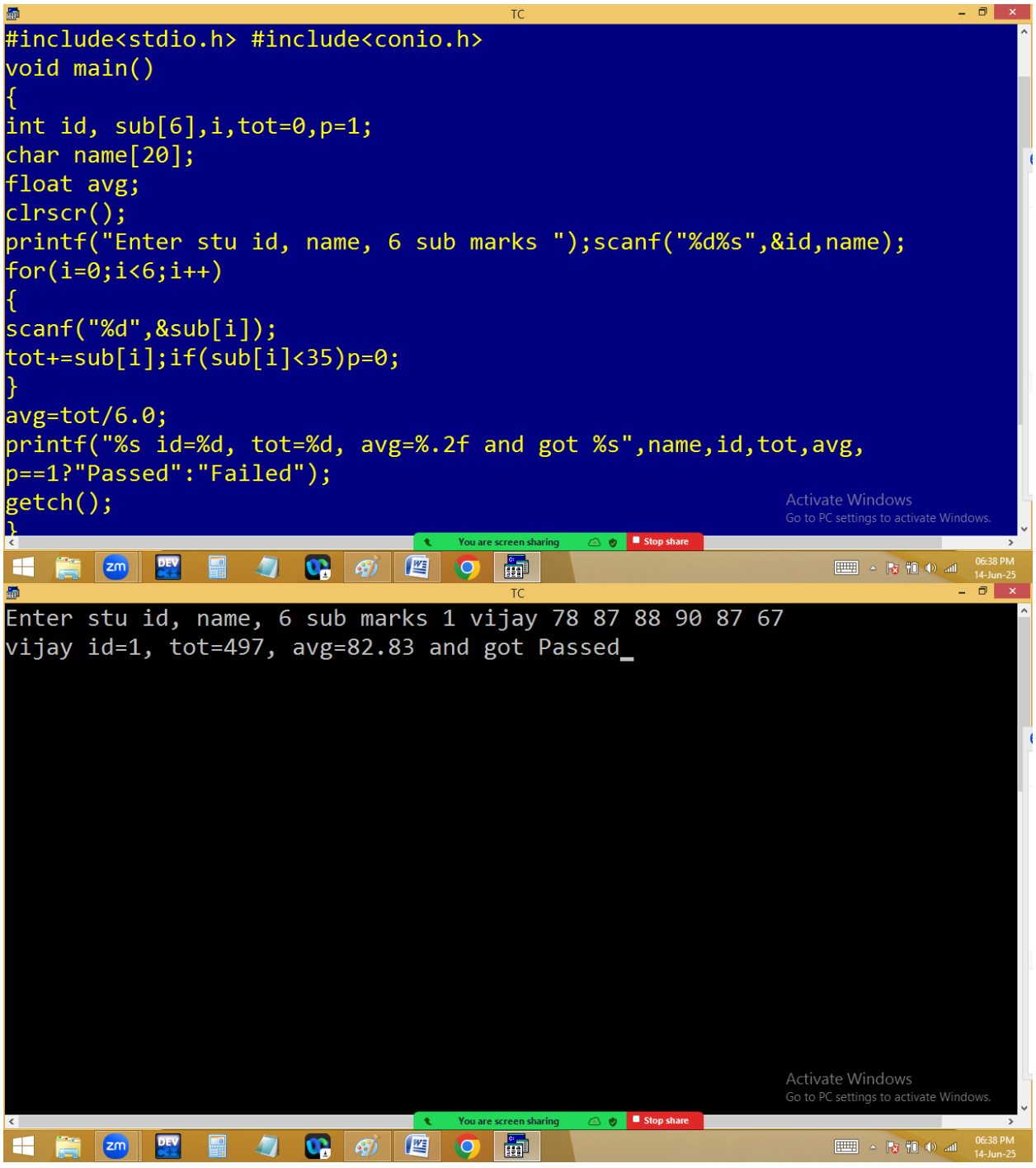
Descending order:

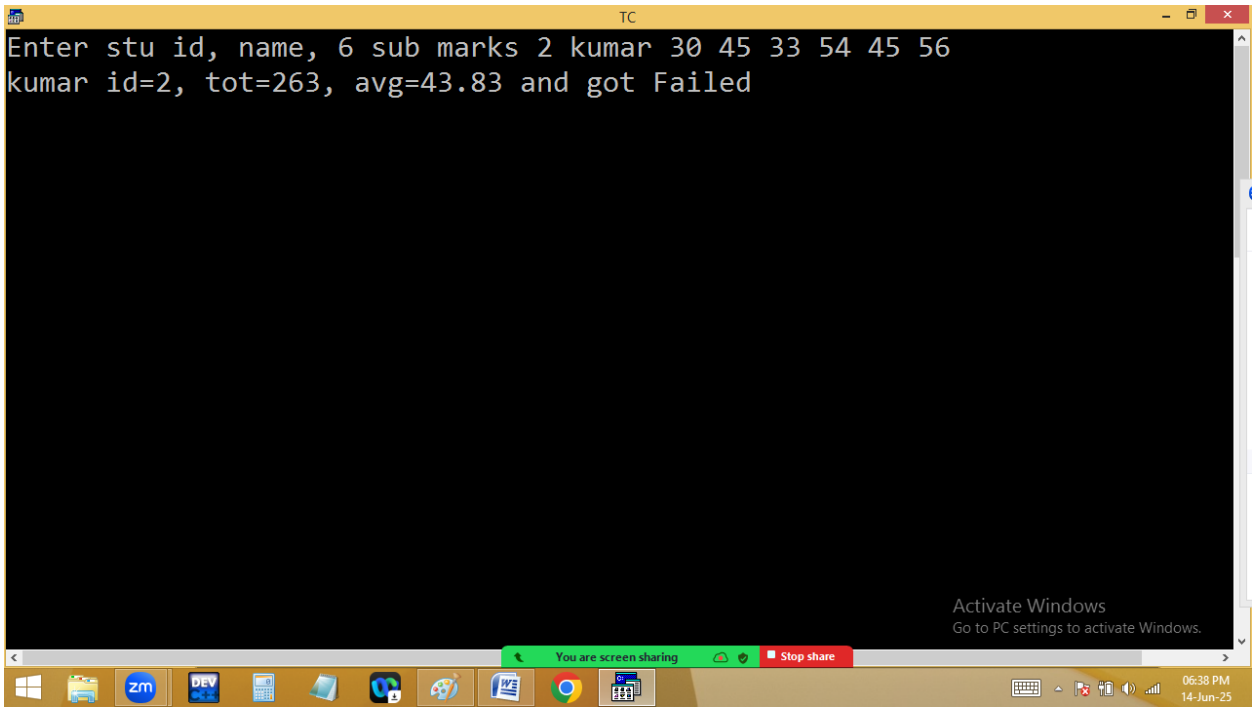


Arrange even elements in ascending order and odd elements order?

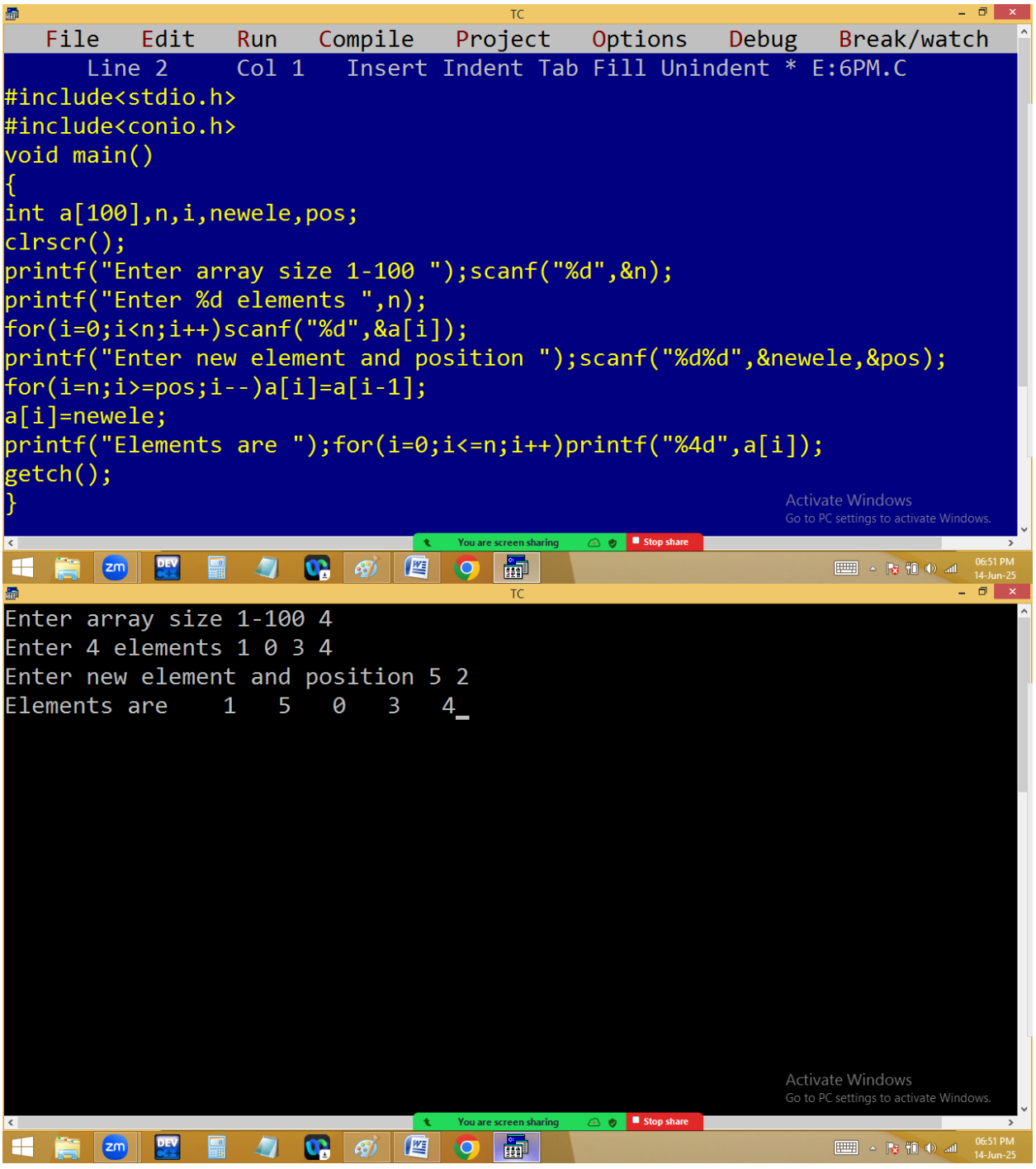


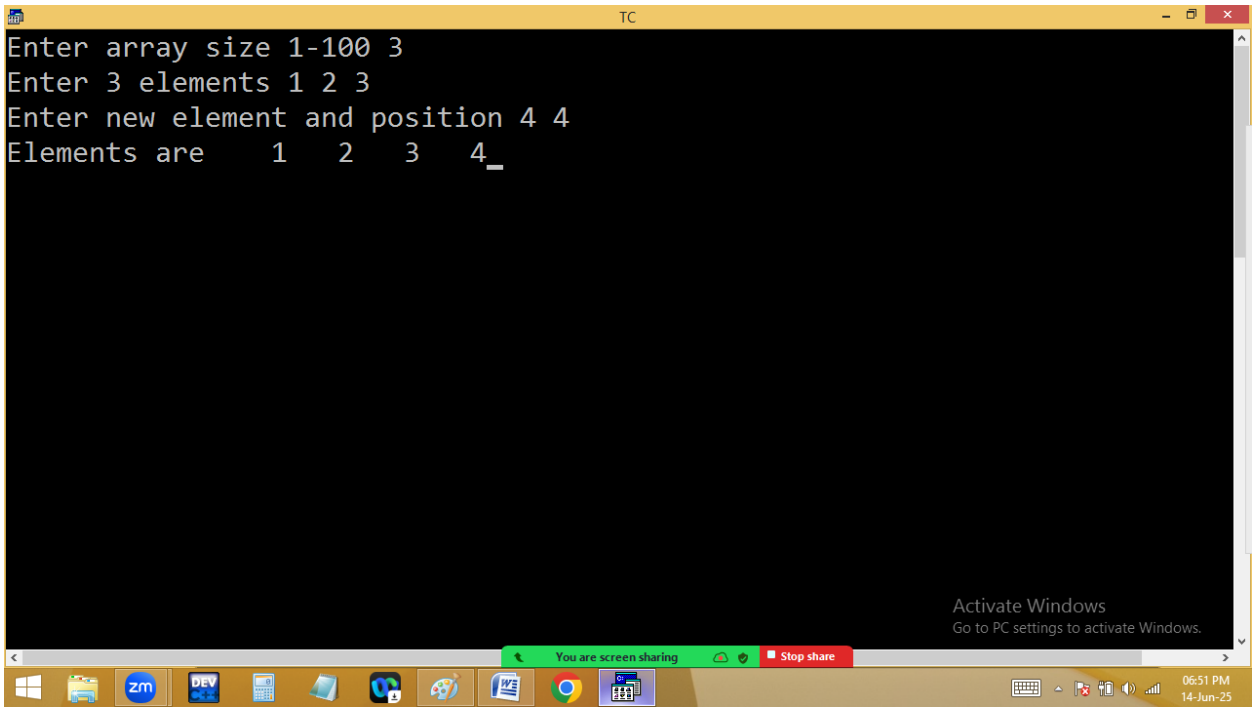
Read a stu id, name, 6 sub marks using array and find the tot, avg and pass / fail.

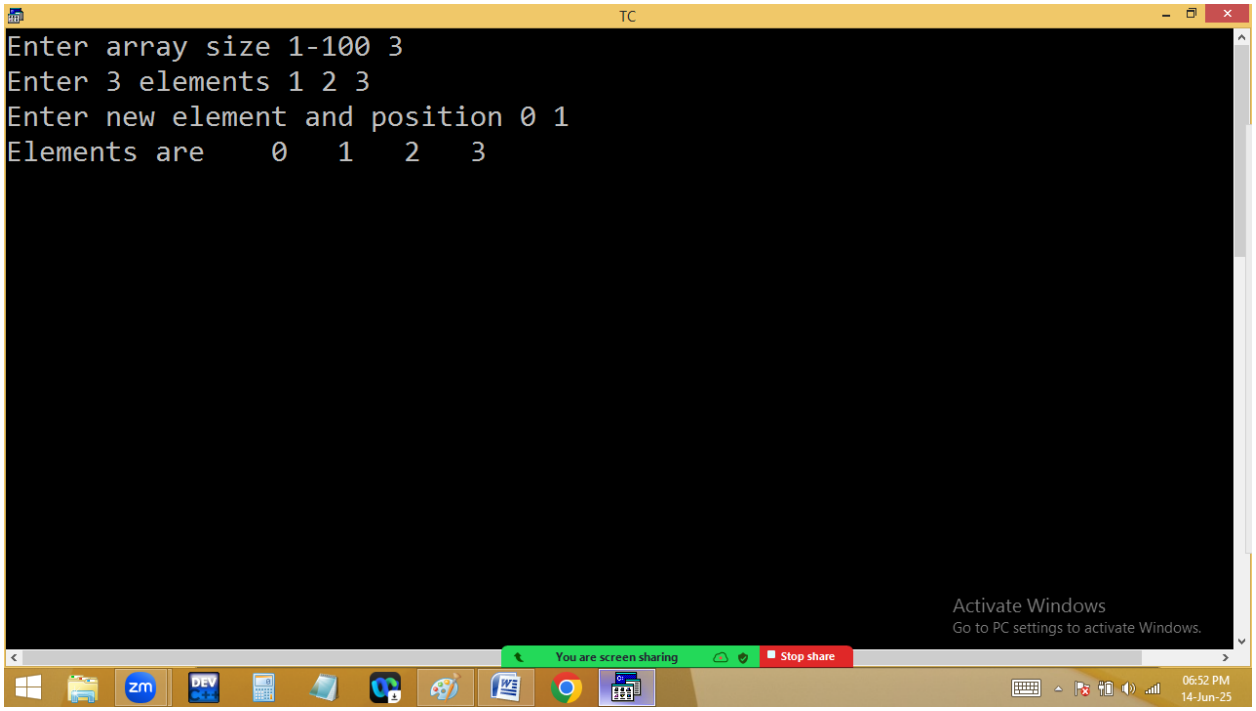

TC
#include<stdio.h> #include<conio.h>
void main()
{
int id, sub[6],i,tot=0,p=1;
char name[20];
float avg;
clrscr();
printf("Enter stu id, name, 6 sub marks ");scanf("%d%s",&id,name);
for(i=0;i<6;i++)
{
scanf("%d",&sub[i]);
tot+=sub[i];if(sub[i]<35)p=0;
}
avg=tot/6.0;
printf("%s id=%d, tot=%d, avg=%.2f and got %s",name,id,tot,avg,
p==1?"Passed":"Failed");
getch();
}
Activate Windows
Go to PC settings to activate Windows.
You are screen sharing
Stop share
06:38 PM
14-Jun-25
TC
Enter stu id, name, 6 sub marks 1 vijay 78 87 88 90 87 67
vijay id=1, tot=497, avg=82.83 and got Passed
Activate Windows
Go to PC settings to activate Windows.
You are screen sharing
Stop share
06:38 PM
14-Jun-25

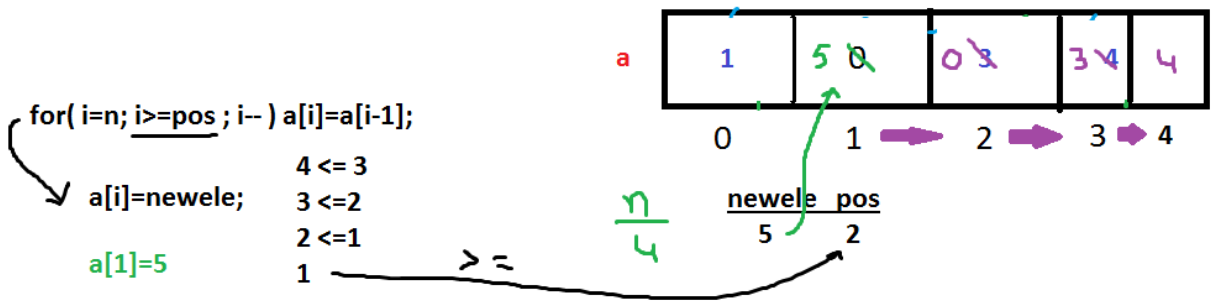


Inserting a new element into the array at specified position [push() or [right shifting of array elements]]?

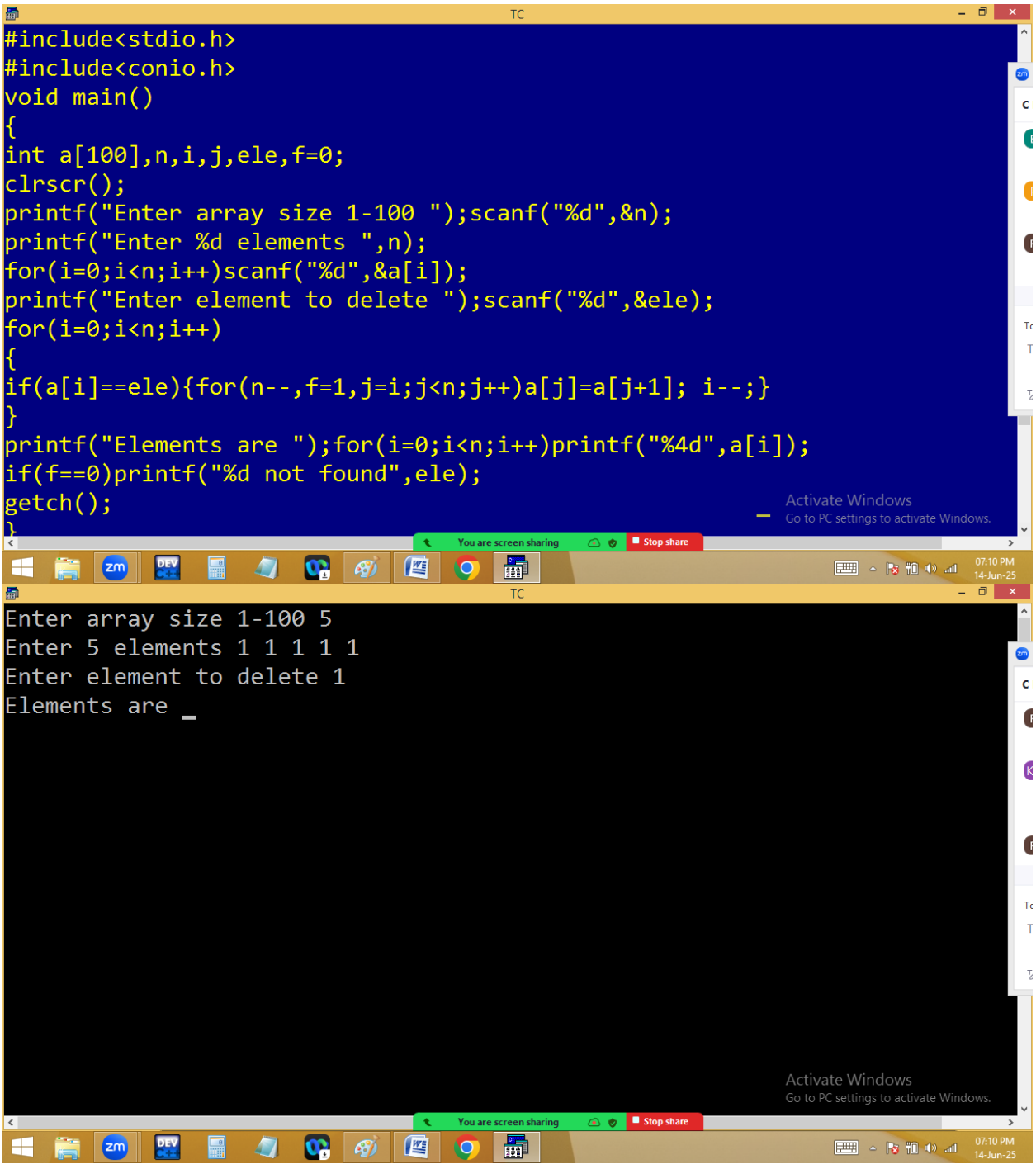

TC
File Edit Run Compile Project Options Debug Break/watch
Line 2 Col 1 Insert Indent Tab Fill Unindent * E:6PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a[100],n,i,newele,pos;
clrscr();
printf("Enter array size 1-100 ");scanf("%d",&n);
printf("Enter %d elements ",n);
for(i=0;i<n;i++)scanf("%d",&a[i]);
printf("Enter new element and position ");scanf("%d%d",&newele,&pos);
for(i=n;i>=pos;i--)a[i]=a[i-1];
a[i]=newele;
printf("Elements are ");for(i=0;i<=n;i++)printf("%4d",a[i]);
getch();
}
Activate Windows
Go to PC settings to activate Windows.
You are screen sharing
Stop share
06:51 PM
14-Jun-25
TC
Enter array size 1-100 4
Enter 4 elements 1 0 3 4
Enter new element and position 5 2
Elements are 1 5 0 3 4
Activate Windows
Go to PC settings to activate Windows.
You are screen sharing
Stop share
06:51 PM
14-Jun-25


TC
Enter array size 1-100 3
Enter 3 elements 1 2 3
Enter new element and position 4 4
Elements are 1 2 3 4
Activate Windows
Go to PC settings to activate Windows.
You are screen sharing
Stop share
06:51 PM
14-Jun-25


TC
Enter array size 1-100 3
Enter 3 elements 1 2 3
Enter new element and position 0 1
Elements are 0 1 2 3
Activate Windows
Go to PC settings to activate Windows.
You are screen sharing
Stop share
06:52 PM
14-Jun-25



Deleting a particular element from array [pop / left shifting of array elements]:


TC
#include<stdio.h>
#include<conio.h>
void main()
{
int a[100],n,i,j,ele,f=0;
clrscr();
printf("Enter array size 1-100 ");scanf("%d",&n);
printf("Enter %d elements ",n);
for(i=0;i<n;i++)scanf("%d",&a[i]);
printf("Enter element to delete ");scanf("%d",&ele);
for(i=0;i<n;i++)
{
if(a[i]==ele){for(n--,f=1,j=i;j<n;j++)a[j]=a[j+1]; i--;}
}
printf("Elements are ");for(i=0;i<n;i++)printf("%4d",a[i]);
if(f==0)printf("%d not found",ele);
getch();
}
Activate Windows
Go to PC settings to activate Windows.
You are screen sharing
Stop share
07:10 PM
14-Jun-25
TC
Enter array size 1-100 5
Enter 5 elements 1 1 1 1 1
Enter element to delete 1
Elements are
Activate Windows
Go to PC settings to activate Windows.
You are screen sharing
Stop share
07:10 PM
14-Jun-25

Home work:

1. **Delete duplicate elements from array.**

Eg: 1 2 3 1 2 3 4 → 1 2 3 4

2. **Frequency of array elements**

Eg: 1 2 3 1 5 1 2

1 found 3 times

2 found 2 times

3 found 1 time

5 found 1 time